

**DATABASE MANAGEMENT SYSTEM
PROJECT
STUDENT RESULT MANAGEMENT SYSTEM**



SUBMITTED TO:

MR ADEEL

SUBMITTED BY:

MALAYKA IBRAR (058)

NATASHA MARYAM(068)

MAIMOONA BIBI (055)

SECTION:

2_B

1. Introduction

The Student Result Management System (SRMS) is designed to efficiently manage and process student academic records and results. Central to its functionality is a robust database that stores, retrieves, and manipulates data related to students, courses, grades, and administrative information.

Explanation of Student Result Management System:

A Student Result Management System (SRMS) is a comprehensive software solution designed to efficiently manage and process academic results for educational institutions. It facilitates the seamless handling of student data, examination records, grading, and generation of reports. Below is a breakdown of its attributes and relationships presented in a tabular form, ensuring clarity.

Entities and attributes:

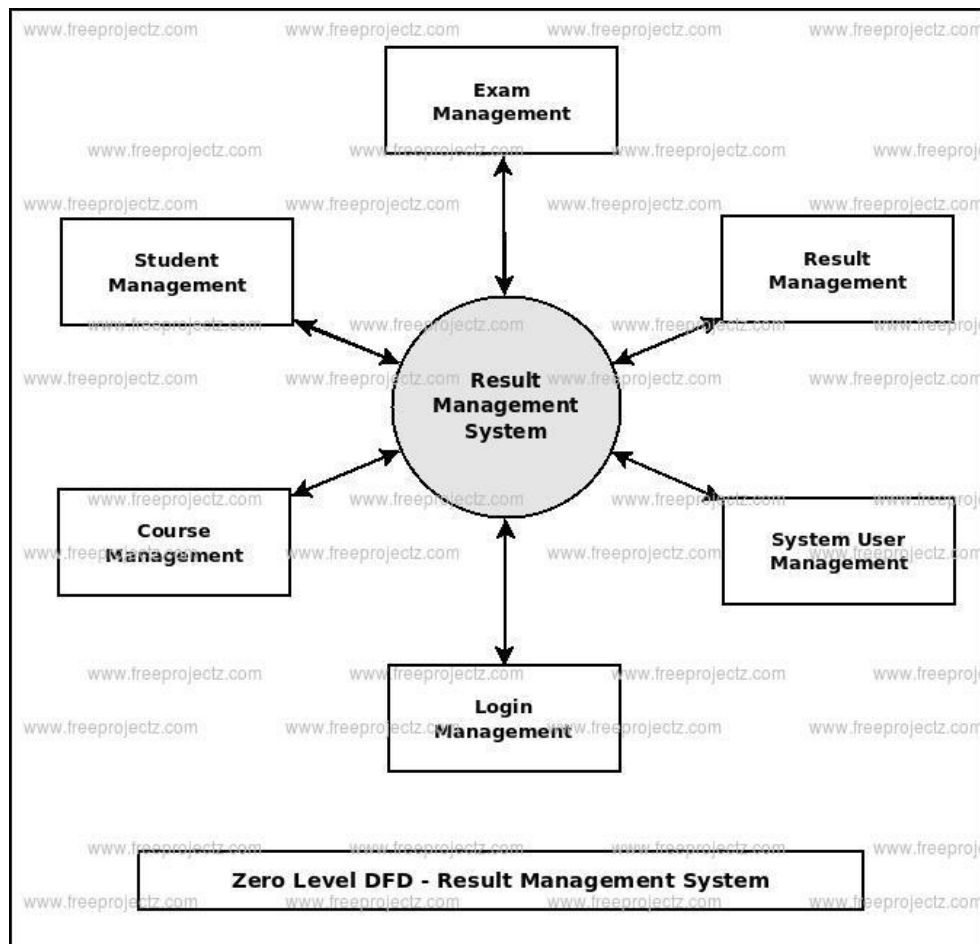
Entity	Attributes
❖ Student	Student ID, Name, Date of Birth, Gender, Contact Information.
❖ Course	Course ID, Course Name
❖ Subject	Subject ID, Subject Name
❖ Examination	Examination ID, Subject ID, Date
❖ Result	Result ID, Student ID, Subject ID, Marks Obtained, Grades.
❖ Faculty	Faculty ID, Faculty Name
❖ NewStudentResult	ResultID, StudentID, SubjectID, MarksObtained

Relationships:

❖ Student - Result	One-to-Many	One student can have multiple results (exams), but each result belongs to only one student.
❖ Course - Subject	One-to-Many	One course can have multiple subjects, but each subject belongs to only one course.
❖ Subject - Result	One-to-Many	One subject can have multiple results (exams), but each result belongs to only one subject.

❖ Student - Course	Many-to-Many	One student can be enrolled in multiple courses, and each course can have multiple students.
❖ Examination - Subject	Many-to-Many	One examination can include multiple subjects, and each subject can be part of multiple examinations.
❖ Faculty - Subject	One-to-Many	One faculty member can teach multiple subjects, but each subject is taught by only one faculty member.
NewStudentResult, Subject	One_to_many	Each new student result is associated with one subject, but a subject can have multiple results.

ERD DIAGRAM OF STUDENT RESULT MANAGEMENT SYSTEM



Tables with inserted data:

CREATE TABLE Student (

StudentID INT PRIMARY KEY,

Name VARCHAR(50),

DateOfBirth DATE,

Gender VARCHAR(10));

INSERT INTO Student (StudentID, Name, DateOfBirth, Gender)

VALUES

(1, 'Ali', '2000-05-15', 'Male'),

(2, 'Fatima', '2001-09-23', 'Female'),

(3, 'Ahmed', '2002-02-10', 'Male'),

(4, 'Aisha', '2000-12-03', 'Female'),

(5, 'Zainab', '2001-07-28', 'Female');

CREATE TABLE NewStudentResult (

ResultID INT PRIMARY KEY,

StudentID INT,

SubjectID INT,

MarksObtained INT,

Grade VARCHAR(10),

ResultStatus VARCHAR(20),

FOREIGN KEY (StudentID) REFERENCES Student(StudentID),

FOREIGN KEY (SubjectID) REFERENCES Subject(SubjectID));

INSERT INTO NewStudentResult (ResultID, StudentID, SubjectID, MarksObtained, Grade, ResultStatus)

VALUES

(201, 1, 101, 85, 'A', 'Pass'),

(202, 2, 102, 78, 'B', 'Pass'),

(203, 3, 103, 92, 'A+', 'Pass'),

(204, 4, 101, 65, 'B-', 'Pass'),

(205, 5, 102, 70, 'B', 'Pass');

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CREATE TABLE Course (

CourseID INT PRIMARY KEY,

Name VARCHAR(50),

Description VARCHAR(255));

INSERT INTO Course (CourseID, Name, Description)

VALUES

(501, 'Computer Science', 'Introduction to programming and computer systems'),

(502, 'Biology', 'Study of living organisms and their interactions'),

(503, 'Economics', 'Study of production, distribution, and consumption of goods and services');

CREATE TABLE Result (

ResultID INT PRIMARY KEY,

ExamID INT,

StudentID INT,

MarksObtained INT,

Grade VARCHAR(10),

ResultStatus VARCHAR(20),

FOREIGN KEY (ExamID) REFERENCES Examination(ExamID),

FOREIGN KEY (StudentID) REFERENCES Student(StudentID));

INSERT INTO Result (ResultID, ExamID, StudentID, MarksObtained, Grade, ResultStatus)

VALUES

(401, 301, 1, 85, 'A', 'Pass'),

(402, 302, 2, 78, 'B', 'Pass'),

(403, 303, 3, 92, 'A+', 'Pass'),

(404, 301, 4, 65, 'B-', 'Pass'),

(405, 302, 5, 70, 'B', 'Pass');

SELECT * FROM Course;

SELECT * FROM Result;

SELECT * FROM Student;

SELECT * FROM NewStudentResult;

Screenshots:

▼  Tables

- ▶  course
- ▶  examination
- ▶  faculty
- ▶  newstudentresult
- ▶  result
- ▶  student
- ▶  subject

CourseID	Name	Description
501	Computer Science	Introduction to programming and computer syst...
502	Biology	Study of living organisms and their interactions
503	Economics	Study of production, distribution, and consumpt...
NULL	NULL	NULL

StudentID	Name	DateOfBirth	Gender
1	Ali	2000-05-15	Male
2	Fatima	2001-09-23	Female
3	Ahmed	2002-02-10	Male
4	Aisha	2000-12-03	Female
5	Zainab	2001-07-28	Female

ExamID	SubjectID	ExamDate
301	101	2024-04-10
302	102	2024-04-12
303	103	2024-04-15
NULL	NULL	NULL

FacultyID	Name	SubjectID
601	Dr. Kareem	101
602	Prof. Amira	102
603	Dr. Ahmed	103
NULL	NULL	NULL

ResultID	ExamID	StudentID	MarksObtained	Grade	ResultStatus
401	301	1	85	A	Pass
402	302	2	78	B	Pass
403	303	3	92	A+	Pass
404	301	4	65	B-	Pass
405	302	5	70	B	Pass

ResultID	StudentID	SubjectID	MarksObtained	Grade	ResultStatus
201	1	101	85	A	Pass
202	2	102	78	B	Pass
203	3	103	92	A+	Pass
204	4	101	65	B-	Pass
205	5	102	70	B	Pass
NULL	NULL	NULL	NULL	NULL	NULL

SubjectID	Name	Description
101	Mathematics	Basic arithmetic and algebra
102	Science	Introduction to biology, chemistry, and physics
103	English	Language and literature
NULL	NULL	NULL

Functionalities

- **Adding and Updating Records:** Allow administrators to add new students, courses, faculty, and departments, and update existing records.
 - **Recording Results:** Facilitate the entry of grades and results for each student per course.
 - **Generating Reports:** Generate reports such as student transcripts, department-wise results, and course performance summaries.
 - **User Access Control:** Implement user roles (e.g., admin, faculty, student) with appropriate permissions to ensure data security and integrity.
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5. Conclusion

The SRMS database design ensures efficient data management, retrieval, and manipulation, supporting the smooth functioning of academic record keeping and result processing. By adhering to relational database principles and normalization, the system enhances reliability, scalability, and maintainability.